Chapter 10. Regression Diagnostics

Hat matrix and leverage, residuals, deletion formulas, influence measures

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10.1 What can go wrong

Fitting $\mathbf{y} = X\beta + \epsilon$, $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, by $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ is justified only if the model is adequate. **Regression diagnostics** look for:

(1) violations of the error assumptions (normality, independence, equal variance), studied through residuals;

(2) a wrong functional form, remedied by transforming y or the x_j (Chapters 5 and 11);

(3) near-singularity of $X'X$ (multicollinearity, Chapter 15);

(4) **high-leverage points**: observations whose $\mathbf{x}_i$ is far from the bulk of the design;

(5) **outliers**: observations whose y_i is far from its fitted value;

(6) **influential observations**: observations whose deletion changes the fit substantially.

Points of types (4)–(6) need not coincide: a high-leverage point can lie on the line, and an outlier at the centre of the design may hardly move the fit.

10.2 The hat matrix and leverage

Motivation. A point far from the others in x-space can pull the fitted surface toward itself. The diagonal of the hat matrix measures this leverage and flags such points before we look at y.

The fitted values are $\hat{\mathbf{y}} = H\mathbf{y}$ with $H = X(X'X)^{-1}X' = (h_{ij})$, where $h_{ij} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_j$ and $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$ is the i-th row of X.

Theorem 10.1 (Properties of the hat matrix)

(a) H is symmetric and idempotent with $\operatorname{rank} H = \operatorname{tr} H = \sum_i h_{ii} = p + 1$.

(b) $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$; hence $0 \le h_{ii} \le 1$ and $|h_{ij}| \le \frac{1}{2}$ for $i \neq j$.

(c) $HX = X$. If the model has an intercept, $H\mathbf{1} = \mathbf{1}$, i.e. every row of H sums to 1, and $h_{ii} \ge 1/n$.

(d) With an intercept, $h_{ii} = \frac{1}{n} + (\mathbf{x}_i^* - \bar{\mathbf{x}})'(\tilde{X}'\tilde{X})^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$, where $\mathbf{x}_i^* = (x_{i1}, \ldots, x_{ip})'$ and $\tilde{X}$ is the centred $n \times p$ matrix of regressors. For $p = 1$, $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$.

Proof. (a) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p + 1$.

(b) The (i,i) entry of $H = HH'$ is $\sum_j h_{ij}^2$. Since it is at least h_{ii}^2, $h_{ii} \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. For $j \neq i$, $h_{ij}^2 \le h_{ii} - h_{ii}^2 \le \frac{1}{4}$.

(c) $HX = X$ is direct; the intercept column gives $H\mathbf{1} = \mathbf{1}$. For $h_{ii} \ge 1/n$ see (d).

(d) Reparametrize with centred regressors: the column space of X is spanned by $\mathbf{1}$ and the columns of $\tilde{X}$, which are orthogonal to $\mathbf{1}$. Hence $H = \frac{1}{n}\mathbf{1}\mathbf{1}' + \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}'$, and the diagonal gives the formula, the second term being nonnegative. For $p = 1$, $\tilde{X}'\tilde{X} = S_{xx}$. ∎

Since $\hat{y}_i = \sum_j h_{ij} y_j$, the weight of y_i in its own fitted value is $h_{ii} = \partial \hat{y}_i / \partial y_i$; h_{ii} is the **leverage** of observation i. By (d), it grows with the (Mahalanobis) distance of $\mathbf{x}_i^*$ from the centroid $\bar{\mathbf{x}}$: with $M(i) = (\mathbf{x}_i^* - \bar{\mathbf{x}})'\{\tilde{X}'\tilde{X}/(n-1)\}^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$,

$$M(i) = (n-1)\Big(h_{ii} - \frac{1}{n}\Big).$$

The average leverage is $\bar{h} = (p+1)/n$, and a common rule flags $h_{ii} > 2\bar{h}$ (or $3\bar{h}$) as high leverage.

10.3 Residuals

Motivation. Raw residuals have unequal variances, so their sizes are not comparable. Scaling them by their standard errors, and deleting the point in question, gives residuals that can be compared with t quantiles to detect outliers.

Theorem 10.2

$\mathbf{e} = (I-H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I-H))$. So $\operatorname{Var} e_i = \sigma^2(1-h_{ii})$, $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$, and $\operatorname{Corr}(e_i, e_j) = -h_{ij}/\sqrt{(1-h_{ii})(1-h_{jj})}$.

High-leverage points have small residual variance: the fit is pulled towards them. To compare residuals on a common scale:

Definition (Scaled residuals)

With $s^2 = MSE = \mathbf{e}'\mathbf{e}/(n-p-1)$ and $s^2_{(i)}$ the MSE computed without observation i:

- the **standardized** (internally studentized) residual is $r_i = \dfrac{e_i}{s\sqrt{1-h_{ii}}}$;
- the **studentized** (externally studentized) residual is $r_i^* = \dfrac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}$.

r_i is not t-distributed, because e_i and s are dependent (a large $|e_i|$ inflates s). Deleting observation i removes the dependence. Let $X_{(i)}$, $\mathbf{y}_{(i)}$ denote X, $\mathbf{y}$ without row i, $\hat\beta_{(i)}$ the corresponding estimator, and $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)}$ the prediction of y_i from the other observations.

Theorem 10.3 (Deletion formulas)

(a) $\hat\beta_{(i)} = \hat\beta - \dfrac{(X'X)^{-1}\mathbf{x}_i e_i}{1-h_{ii}}$.

(b) $y_i - \tilde y_i = \dfrac{e_i}{1-h_{ii}}$ (the **deleted** or PRESS residual).

(c) $(n-p-2)s^2_{(i)} = (n-p-1)s^2 - \dfrac{e_i^2}{1-h_{ii}}$.

Proof [Supplement]. (a) $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. With $A = X'X$, the Sherman–Morrison formula gives

$$(A - \mathbf{x}_i\mathbf{x}_i')^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1-h_{ii}}.$$

Multiplying by $X'\mathbf{y} - \mathbf{x}_i y_i$ and using $A^{-1}X'\mathbf{y} = \hat\beta$, $\mathbf{x}_i'\hat\beta = \hat y_i$, $\mathbf{x}_i'A^{-1}\mathbf{x}_i = h_{ii}$:

$$\hat\beta_{(i)} = \hat\beta - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat y_i - h_{ii}y_i)}{1-h_{ii}} = \hat\beta + \frac{A^{-1}\mathbf{x}_i\{\hat y_i - h_{ii}y_i - (1-h_{ii})y_i\}}{1-h_{ii}} = \hat\beta - \frac{A^{-1}\mathbf{x}_i e_i}{1-h_{ii}}.$$

(b) $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)} = \hat y_i - h_{ii}e_i/(1-h_{ii})$, so $y_i - \tilde y_i = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$.

(c) For $j \neq i$, the deleted residual is $y_j - \mathbf{x}_j'\hat\beta_{(i)} = e_j + h_{ji}e_i/(1-h_{ii})$. Summing squares over $j \neq i$ and using $\sum_j e_j h_{ji} = 0$ (since $H\mathbf{e} = \mathbf{0}$) and $\sum_j h_{ji}^2 = h_{ii}$:

$$SSE_{(i)} = \sum_{j\neq i} e_j^2 + \frac{2e_i}{1-h_{ii}}\sum_{j\neq i} h_{ji}e_j + \frac{e_i^2}{(1-h_{ii})^2}\sum_{j\neq i} h_{ji}^2.$$

Here $\sum_{j\neq i} e_j^2 = SSE - e_i^2$, $\sum_{j\neq i} h_{ji}e_j = -h_{ii}e_i$, and $\sum_{j\neq i} h_{ji}^2 = h_{ii} - h_{ii}^2$. So

$$SSE_{(i)} = SSE - e_i^2 - \frac{2h_{ii}e_i^2}{1-h_{ii}} + \frac{h_{ii}e_i^2}{1-h_{ii}} = SSE - \frac{e_i^2}{1-h_{ii}}.$$

■

Theorem 10.4 (Distribution of the studentized residual)

(a) $r_i^* = \dfrac{y_i - \tilde{y}_i}{s_{(i)}\sqrt{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i}} \sim t(n-p-2)$.

(b) $r_i^* = r_i\sqrt{\dfrac{n-p-2}{n-p-1-r_i^2}}$, a monotone function of r_i.

Proof. (a) y_i is independent of $\hat{\beta}_{(i)}$ and $s_{(i)}^2$, which are computed from the other $n-1$ observations; so $y_i - \tilde{y}_i \sim N(0, \sigma^2\{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i\})$, independent of $(n-p-2)s_{(i)}^2/\sigma^2 \sim \chi^2(n-p-2)$. By Sherman–Morrison, $1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i = 1 + h_{ii} + h_{ii}^2/(1-h_{ii}) = 1/(1-h_{ii})$, and with Theorem 10.3(b) the ratio equals $e_i/\{s_{(i)}\sqrt{1-h_{ii}}\}$.

(b) Substitute Theorem 10.3(c), $s_{(i)}^2 = s^2(n-p-1-r_i^2)/(n-p-2)$. ■

So an observation is flagged as an outlier when $|r_i^*| > t_{\alpha/2}(n-p-2)$. Since n tests are performed, a Bonferroni level α/n is safer.

10.4 Influence measures

An observation is influential if deleting it changes the fitted model a lot. The measures below combine the residual and the leverage.

Theorem 10.5 (Influence measures)

(a) **DFFITS**: $\mathrm{DFFITS}_{(i)} = \dfrac{\hat{y}_i - \tilde{y}_{i(i)}}{s_{(i)}\sqrt{h_{ii}}} = \left(\dfrac{h_{ii}}{1-h_{ii}}\right)^{1/2} r_i^*$, where $\tilde{y}_{i(i)} = \mathbf{x}_i'\hat{\beta}_{(i)}$.

(b) **Cook's distance**: $D_{(i)} = \dfrac{(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)})}{(p+1)s^2} = \dfrac{r_i^2}{p+1}\cdot\dfrac{h_{ii}}{1-h_{ii}}$.

(c) **COVRATIO**: $\mathrm{COVRATIO}_{(i)} = \dfrac{\det\{s_{(i)}^2(X_{(i)}'X_{(i)})^{-1}\}}{\det\{s^2(X'X)^{-1}\}} = \dfrac{1}{(1-h_{ii})}\left(\dfrac{n-p-2+r_i^{*2}}{n-p-1}\right)^{-(p+1)}$.

(d) **Andrews–Pregibon**: with $Z = (X, \mathbf{y})$, $\mathrm{AP}_{(i)} = \dfrac{\det(Z_{(i)}'Z_{(i)})}{\det(Z'Z)} = 1 - h_{ii} - \dfrac{e_i^2}{SSE}$.

Proof [Supplement]. (a) By Theorem 10.3(a), $\hat{y}_i - \tilde{y}_{i(i)} = \mathbf{x}_i'(\hat{\beta} - \hat{\beta}_{(i)}) = h_{ii}e_i/(1-h_{ii})$; divide by $s_{(i)}\sqrt{h_{ii}}$.

(b) $(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)}) = \frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i = \frac{h_{ii}e_i^2}{(1-h_{ii})^2}$; divide by $(p+1)s^2$.

(c) By the matrix determinant lemma,

$$\det(X_{(i)}'X_{(i)}) = \det(X'X)\,(1-h_{ii}),$$

so the ratio is $(s^2_{(i)}/s^2)^{p+1}/(1-h_{ii})$. By Theorem 10.3(c), $s^2_{(i)}/s^2 = (n-p-1-r_i^2)/(n-p-2)$, and by Theorem 10.4(b), $n-p-2+r_i^{*2} = (n-p-2)(n-p-1)/(n-p-1-r_i^2)$. Hence $s^2_{(i)}/s^2 = (n-p-1)/(n-p-2+r_i^{*2})$, which gives the formula.

(d) By the determinant lemma, $\det(Z'_{(i)}Z_{(i)})/\det(Z'Z) = 1 - \mathbf{z}_i'(Z'Z)^{-1}\mathbf{z}_i$, where $\mathbf{z}_i' = (\mathbf{x}_i', y_i)$. The last quantity is the leverage of observation i in the regression on Z, which by the partitioned-inverse formula equals $h_{ii} + e_i^2/SSE$. ■

Rules of thumb (Belsley, Kuh and Welsch; Cook):

- $|\text{DFFITS}_{(i)}| > 2\sqrt{(p+1)/n}$;
- $D_{(i)} > F_{0.50}(p+1, n-p-1)$ (roughly 1): deleting i moves $\hat\beta$ outside the 50% confidence ellipsoid of Theorem 8.2;
- $|\text{COVRATIO}_{(i)} - 1| > 3(p+1)/n$;
- small $\text{AP}_{(i)}$.

Every measure grows with both $|r_i|$ and h_{ii}: influence is "outlyingness × leverage".

Example 10.1 (Two kinds of unusual point)

In a classical data set of $n = 21$ children (age at first word x, aptitude score y), the fit is $\hat y = 109.87 - 1.13x$. Observation 18 has $x = 42$, far from the others, with $h_{18,18} = 0.65 > 3\bar h = 0.29$: a high-leverage point. Its residual is small ($r^* = -0.85$), yet $D_{(18)} = 0.68$ and $\text{DFFITS}_{(18)} = -1.16$ are the largest: it is influential. Observation 19 has the largest residual, $r^*_{19} = 3.61 > t_{0.025}(18) = 2.10$: an outlier, but with ordinary leverage ($h = 0.053$), so its Cook's distance is only 0.22. The two points call for different actions: check the measurement of the outlier, and assess whether the conclusions depend on the single high-leverage point.

What to do. An unusual point is first checked for recording errors. If it is genuine, it may signal a wrong model (a missing variable or nonlinearity); robust regression or reporting fits with and without the point are better than silently deleting it. Influence diagnostics based on deleting one point at a time can miss groups of points that mask each other.

10.5 Selected exercises

Motivation. The deletion formulas answer concrete questions: what happens when a point lies exactly on the fitted surface, what a leverage of one means, and why the standardized residual cannot have a t distribution.

Exercise 10.1 (Deleting a point on the fitted surface)

Suppose $y_i = \hat y_i$ for some i, and that $X_{(i)}$ still has full column rank. Show directly from the normal equations that deleting observation i does not change the least squares estimator, and find $s^2_{(i)}$ in terms of s^2.

Solution.

Step 1. $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. Hence, for the full-data estimator $\hat\beta$,

$$X'_{(i)}(\mathbf{y}_{(i)} - X_{(i)}\hat\beta) = X'(\mathbf{y} - X\hat\beta) - \mathbf{x}_i(y_i - \mathbf{x}_i'\hat\beta) = \mathbf{0} - \mathbf{x}_i e_i = \mathbf{0},$$

since $X'\mathbf{e} = \mathbf{0}$ and $e_i = 0$.

Step 2. So $\hat{\beta}$ solves the normal equations of the reduced data. As $X'_{(i)}X_{(i)}$ is nonsingular, the solution is unique: $\hat{\beta}_{(i)} = \hat{\beta}$. (This is the case $e_i = 0$ of Theorem 10.3(a).)

Step 3. The remaining residuals are unchanged, so $SSE_{(i)} = \sum_{j\neq i} e_j^2 = SSE$, now with $n-p-2$ degrees of freedom: $s^2_{(i)} = \frac{n-p-1}{n-p-2}s^2$, which is larger than s^2 when $SSE > 0$.

Such a point has zero Cook's distance however large its leverage, illustrating that leverage alone does not make a point influential. ■

Exercise 10.2 (Leverage equal to one)

Suppose $h_{ii} = 1$.
(a) Show that $h_{ij} = 0$ for $j \neq i$, and that $\hat{y}_i = y_i$ and $e_i = 0$ for every response vector $\mathbf{y}$.
(b) Show that $X'_{(i)}X_{(i)}$ is singular, so that β cannot be estimated without observation i; conversely, if $X_{(i)}$ has full column rank, then $h_{ii} < 1$.
(c) In simple regression with $x = (0, \ldots, 0, 1)$ (with $n \geq 2$), compute h_{nn}.

Solution [Supplement].

(a) By Theorem 10.1(b), $h_{ii} = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2$, i.e. $1 = 1 + \sum_{j\neq i} h_{ij}^2$, so every $h_{ij} = 0$ $(j \neq i)$. Then $\hat{y}_i = \sum_j h_{ij}y_j = y_i$ and $e_i = 0$, whatever $\mathbf{y}$ is.

(b) By the matrix determinant lemma (as in the proof of Theorem 10.5(c)),

$$\det(X'_{(i)}X_{(i)}) = \det(X'X - \mathbf{x}_i\mathbf{x}_i') = \det(X'X)\,\{1 - \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i\} = \det(X'X)(1 - h_{ii}).$$

So $h_{ii} = 1$ makes $X'_{(i)}X_{(i)}$ singular, i.e. $X_{(i)}$ rank deficient. Conversely, if $X_{(i)}$ has full column rank, the determinant is positive, so $h_{ii} \neq 1$, and $h_{ii} < 1$ by Theorem 10.1(b).

(c) $\bar{x} = 1/n$ and $S_{xx} = (n-1)/n^2 + (1-1/n)^2 = (n-1)/n$. By Theorem 10.1(d),

$$h_{nn} = \frac{1}{n} + \frac{(1-1/n)^2}{(n-1)/n} = \frac{1}{n} + \frac{n-1}{n} = 1.$$

The other observations all have $x = 0$, so they carry no information about the slope: the single point at $x = 1$ determines the slope alone, and the line passes through it. ■

Exercise 10.3 (Standardized residuals are bounded)

Assume $h_{ii} < 1$.
(a) Show that $r_i^2 \leq n-p-1$. Conclude that r_i does not have a t distribution.
(b) Show that the map $r_i \mapsto r_i^*$ of Theorem 10.4(b) is a strictly increasing bijection from $(-\sqrt{n-p-1}, \sqrt{n-p-1})$ onto $\mathbb{R}$.

Solution [Supplement].

(a) By Theorem 10.3(c), $0 \leq (n-p-2)s^2_{(i)} = (n-p-1)s^2 - e_i^2/(1-h_{ii})$. Hence $e_i^2/\{s^2(1-h_{ii})\} \leq n-p-1$, i.e. $r_i^2 \leq n-p-1$, with equality iff $SSE_{(i)} = 0$. A t variable takes arbitrarily large values with positive probability, so r_i cannot be t-distributed.

(b) On $|r| < \sqrt{n-p-1}$ let $f(r) = r\sqrt{(n-p-2)/(n-p-1-r^2)}$. It is continuous and odd. For $0 \leq r_1 < r_2 < \sqrt{n-p-1}$, the factor r increases and the factor $(n-p-1-r^2)^{-1/2}$ increases, both being positive for $r > 0$, so $f(r_1) < f(r_2)$; by oddness f is strictly increasing on the whole interval. As $r \uparrow \sqrt{n-p-1}$, $f(r) \to +\infty$, and $f(r) \to -\infty$ as $r \downarrow -\sqrt{n-p-1}$. By the intermediate value theorem f is onto $\mathbb{R}$. So r_i^* and r_i rank the observations identically, but only r_i^* has the unbounded $t(n-p-2)$ distribution. ■